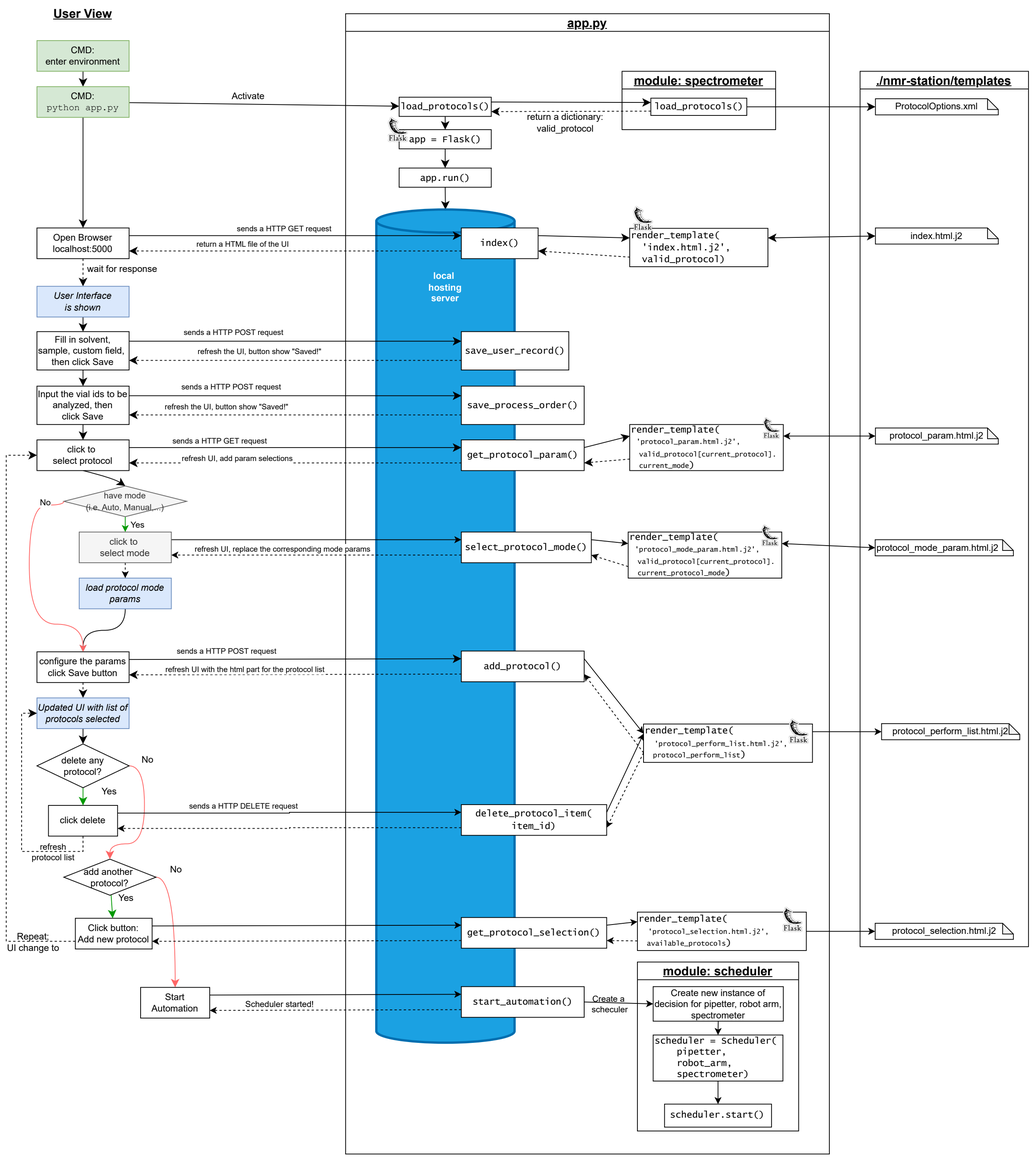
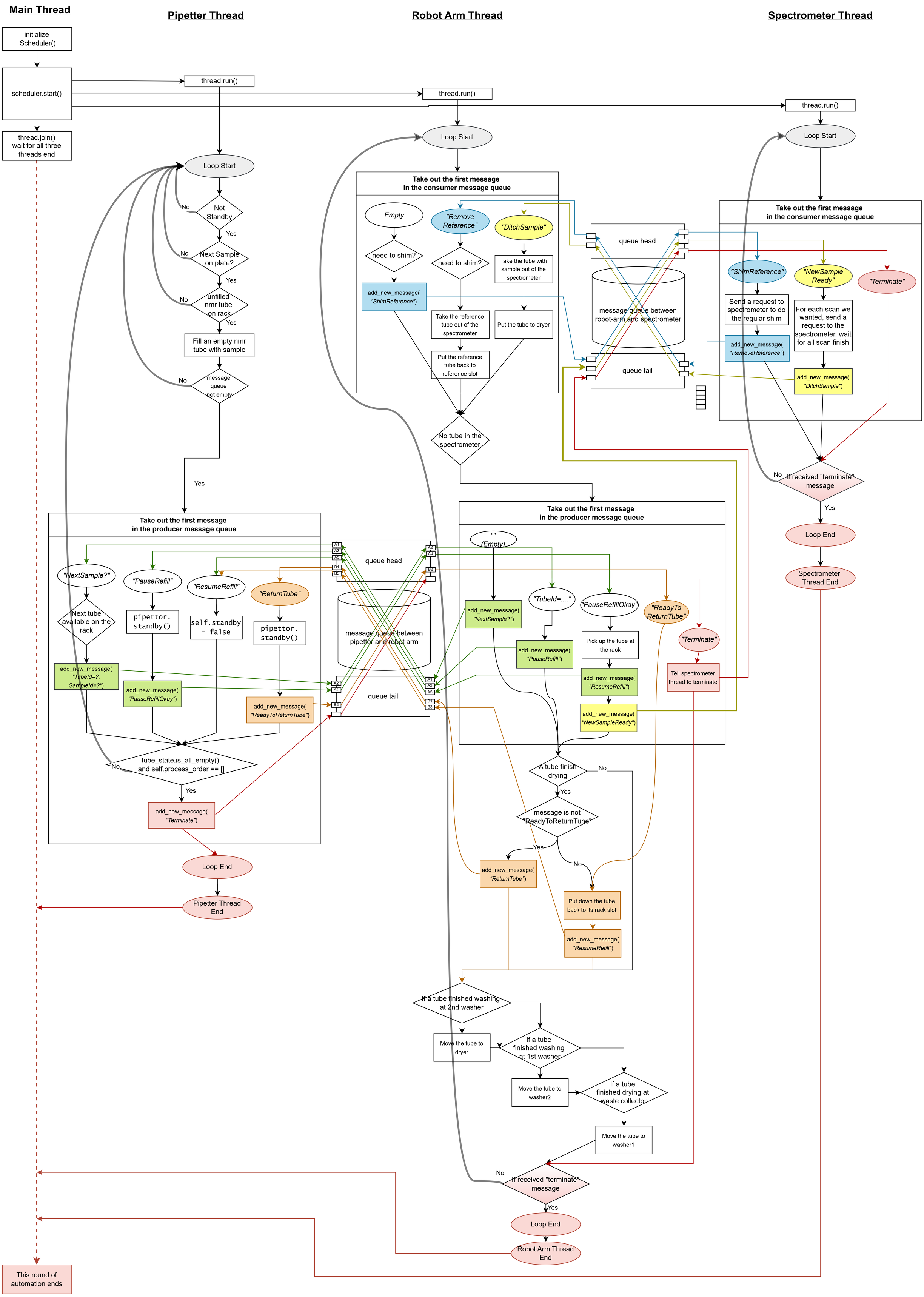
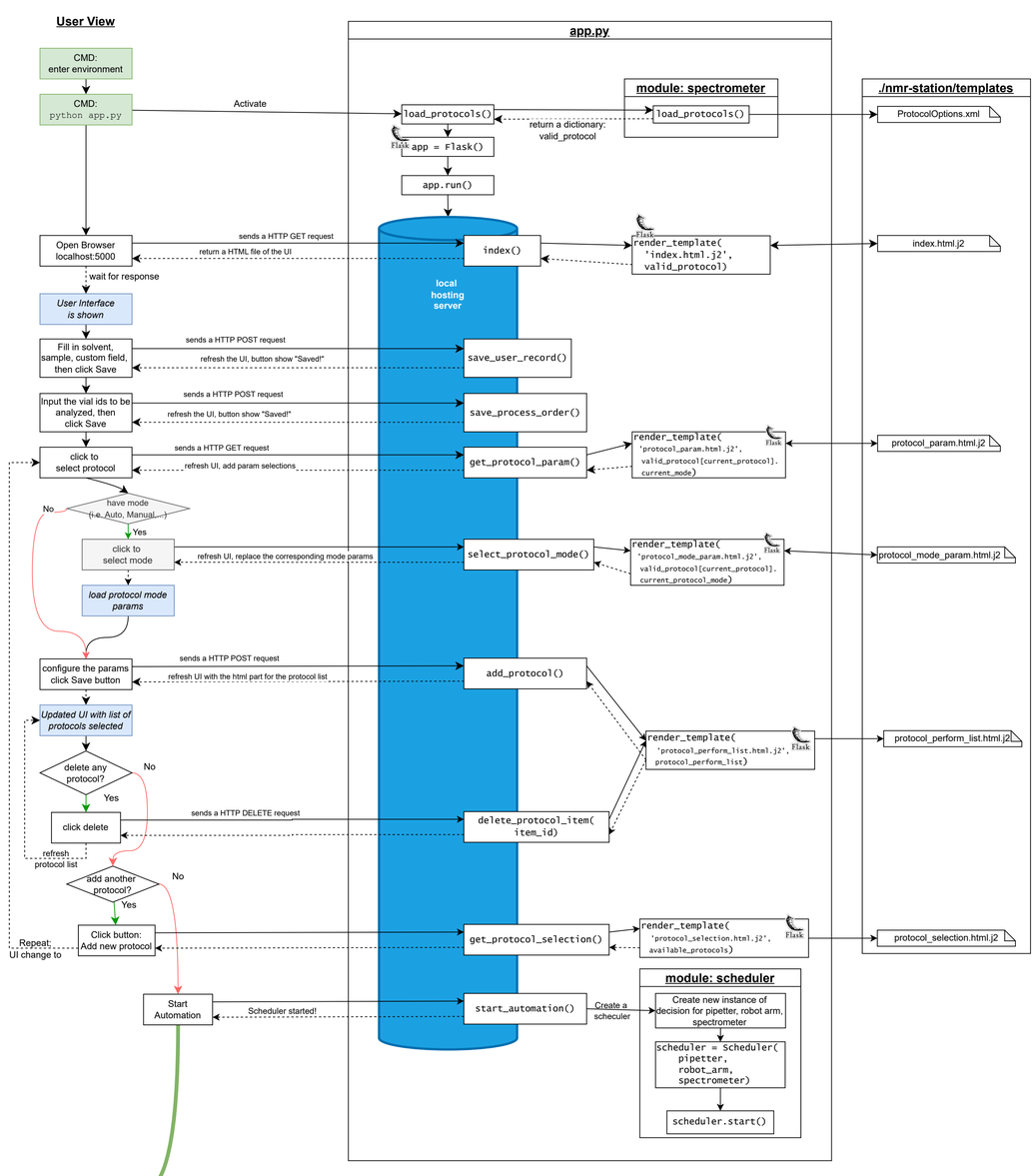

User View
app.py
module: spectrometer
./nmr-station/templates
CMD:
enter environment
CMD:
python app.py
Activate
load_protocols()
load_protocols()
return a dictionary:
valid_protocol
ProtocolOptions.xml
app = Flask()
app.run()
Open Browser
localhost:5000
sends a HTTP GET request
return a HTML file of the UI
index()
render_template(
'index.html.j2',
valid_protocol)
index.html.j2
wait for response
User Interface
is shown
local
hosting
server
Fill in solvent,
sample, custom field,
then click Save
sends a HTTP POST request
refresh the UI, button show "Saved!"
save_user_record()
Input the vial ids to be
analyzed, then
click Save
sends a HTTP POST request
refresh the UI, button show "Saved!"
save_process_order()
click to
select protocol
sends a HTTP GET request
refresh UI, add param selections
get_protocol_param()
render_template(
'protocol_param.html.j2',
valid_protocol[current_protocol].
current_mode)
protocol_param.html.j2
have mode
(i.e. Auto, Manual...)
No
Yes
click to
select mode
refresh UI, replace the corresponding mode params
select_protocol_mode()
render_template(
'protocol_mode_param.html.j2',
valid_protocol[current_protocol].
current_protocol_mode)
protocol_mode_param.html.j2
load protocol mode
params
configure the params
click Save button
sends a HTTP POST request
refresh UI with the html part for the protocol list
add_protocol()
Updated UI with list of
protocols selected
render_template(
'protocol_perform_list.html.j2',
protocol_perform_list)
protocol_perform_list.html.j2
delete any
protocol?
No
Yes
click delete
sends a HTTP DELETE request
delete_protocol_item(
item_id)
refresh
protocol list
add another
protocol?
No
Yes
Click button:
Add new protocol
Repeat
UI change to
get_protocol_selection()
render_template(
'protocol_selection.html.j2',
available_protocols)
protocol_selection.html.j2
module: scheduler
Start
Automation
Scheduler started!
start_automation()
Create a
scheculer
Create new instance of
decision for pipetter, robot arm,
spectrometer
scheduler = Scheduler(
pipetter,
robot_arm,
spectrometer)
scheduler.start()


Main Thread
Pipetter Thread
Robot Arm Thread
Spectrometer Thread
initialize Scheduler()
scheduler.start()
thread.join() wait for all three threads end
thread.run()
Loop Start
Not Standby
Next Sample on plate?
unfilled nmr tube on rack
Fill an empty nmr tube with sample
message queue not empty
No
Yes
Take out the first message in the consumer message queue
Empty
need to shim?
add_new_message("ShimReference")
"Remove Reference"
Take the reference tube out of the spectrometer
Put the reference tube back to reference slot
"DitchSample"
Take the tube with sample out of the spectrometer
Put the tube to dryer
No tube in the spectrometer
queue head
queue tail
message queue between robot-arm and spectrometer
"ShimReference"
Send a request to spectrometer to do the regular shim
add_new_message("RemoveReference")
"NewSample Ready"
For each scan we wanted, send a request to the spectrometer, wait for all scan finish
add_new_message("DitchSample")
"Terminate"
If received "terminate" message
Loop End
Spectrometer Thread End
Take out the first message in the producer message queue
"NextSample?"
Next tube available on the rack
add_new_message("TubeId=?, SampleId=?")
"PauseRefill"
pipettor. standby()
add_new_message("PauseRefillOkay")
"ResumeRefill"
self.standby = false
"ReturnTube"
pipettor. standby()
add_new_message("ReadyToReturnTube")
tube_state.is_all_empty() and self.process_order == []
add_new_message("Terminate")
Loop End
Pipetter Thread End
message queue between pipettor and robot arm
queue head
queue tail
""
(Empty)
add_new_message("NextSample?")
"TubeId=...."
add_new_message("PauseRefill")
"PauseRefillOkay"
Pick up the tube at the rack
add_new_message("ResumeRefill")
add_new_message("NewSampleReady")
"ReadyTo ReturnTube"
"Terminate"
Tell spectrometer thread to terminate
A tube finish drying
message is not "ReadyToReturnTube"
add_new_message("ReturnTube")
Put down the tube back to its rack slot
add_new_message("ResumeRefill")
If a tube finished washing at 2nd washer
Move the tube to dryer
If a tube finished washing at 1st washer
Move the tube to washer2
If a tube finished drying at waste collector
Move the tube to washer1
If received "terminate" message
Loop End
Robot Arm Thread End
This round of automation ends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add_protocol()
Updated UI with list of protocols selected
delete any protocol?
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refresh protocol list
add another protocol?
Click button: Add new protocol
Repeat UI change to
get_protocol_selection()
protocol_selection.html.j2
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Scheduler started!
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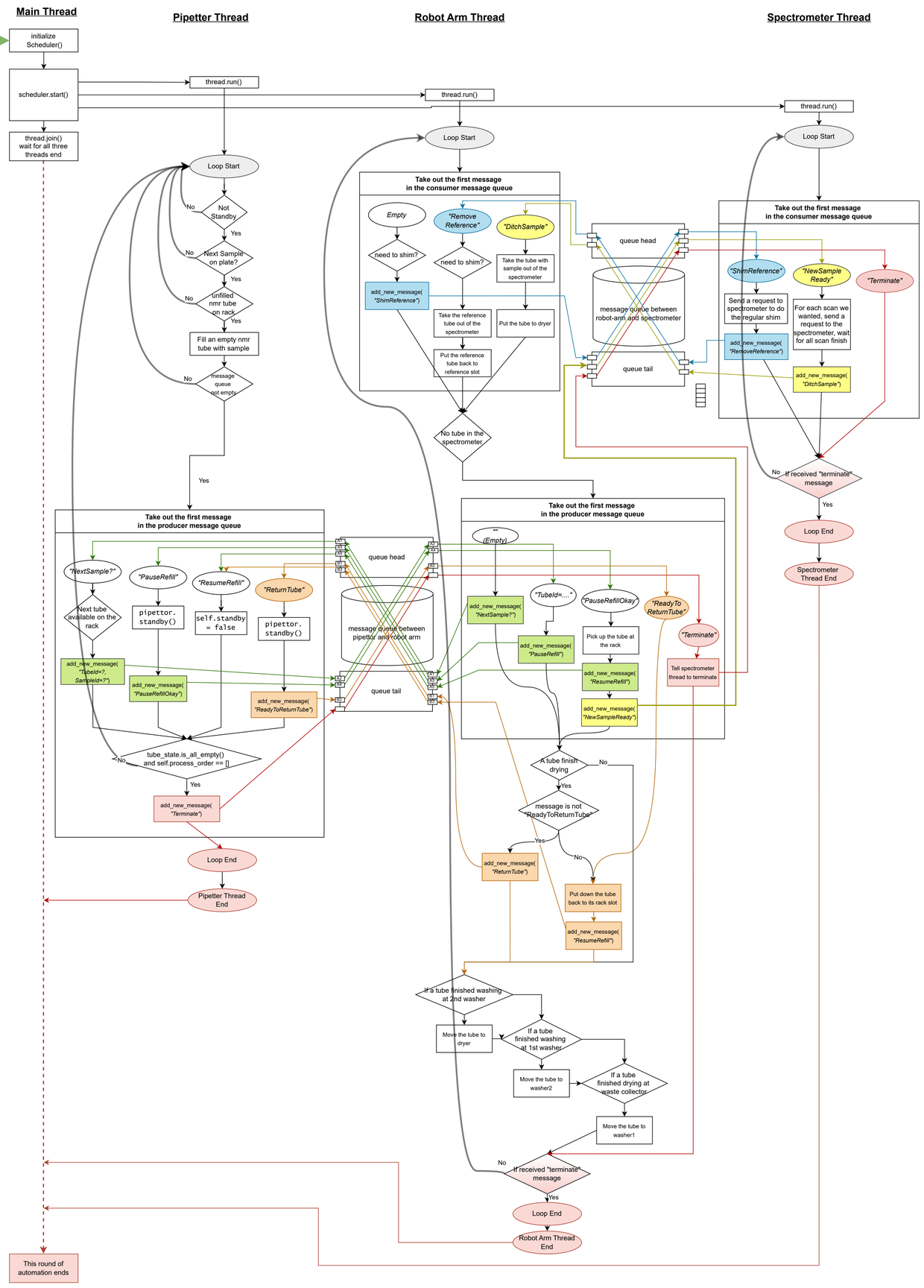
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add_new_message("ReadyToReturnTube")
tube_state.is_all_empty() and self.process_order == []
add_new_message("Terminate")
Pipetter Thread End
message queue between pipettor and robot arm
"(Empty)"
"TubeId=...."
"PauseRefillOkay"
"ReadyTo ReturnTube"
add_new_message("NextSample?")
Pick up the tube at the rack
add_new_message("ResumeRefill")
add_new_message("ViewSampleReady")
Tell spectrometer thread to terminate
A tube finish drying
message is not "ReadyToReturnTube"
Put down the tube back to its rack slot
If a tube finished washing at 2nd washer
Move the tube to dryer
If a tube finished washing at 1st washer
Move the tube to washer2
If a tube finished drying at waste collector
Move the tube to washer1
Robot Arm Thread End
This round of automation ends